

Hobbies for Geeks:

Code Golfing and Code Obfuscation

This article introduces the reader to two activities that are considered hobby programming — code golfing and code obfuscation.

```
#include<stdio.h>
int main()
{
    int f1=0,f2=1,f3,ctr=3;
    printf("%d\n%d\n",f1,f2);
    while(ctr < 32)
    {
        f3=f1+f2;
        f1=f2;
        f2=f3;
        printf("%d\n",f3);
        ctr++;
    }
    return 0;
}
```

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#include<stdio.h>
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```

People sometimes wonder about the hobbies of computer geeks who are often introverts and indoor people.

To many computer programmers, programming is just another profession to make a living, often considered a tiresome task. But there are a few who like to write programs for enjoyment and as a hobby.

What is code golfing?

First, let us try to understand the rules of code golfing by looking at how the game of golf works. To win a game of golf you have to use the minimum number of strokes to hit the ball into all the 18 holes on the course, sequentially. Ironically, the aim of golf is to play for the least amount of time. With that knowledge about the game, let's try to understand what code golfing is all about. In code golfing, a programmer tries to solve a problem with the shortest possible source code. But do remember

that a 'code golfed' program may not be the best in terms of parameters like time complexity, space complexity (different from source code size), readability, etc, which are absolutely important when you are developing software.

Now let us consider an example for code golfing. Let us write a program to print the first 31 Fibonacci numbers from 0 to 832040, each on a separate line. This is a question posed by a code golfing puzzles website (<https://code-golf.io>). I have decided to stick with C, one of my favourite programming languages, for the solution. The program given below, called *fib1.c*, is my first attempt. This and all the other programs discussed in this article can be downloaded from opensourceforu.com/article_source_code/May20HobbiesforGeeks.zip.

```
#include<stdio.h>
int main()
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int f1=0,f2=1,f3,ctr=3;
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while(ctr < 32)
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    f3=f1+f2;
    f1=f2;
    f2=f3;
    printf("%d\n",f3);
    ctr++;
}
return 0;
}
```

This program successfully completed the challenge and the source code has 194 characters (including white space characters). Of course, this size was quite expected, as my code did not use any golfing techniques and was just plain enough to print the Fibonacci numbers. I have worked on this problem for some time and finally came up with the program called *fib2.c* given below. Notice that the program *fib2.c* is written in a single line to save some white space characters.

